



Initramfs for boot time reduction

Create the smallest initramfs possible, just enough to start the critical application and then switch to the final root filesystem with `switch_root`:

- ▶ Use a light C library reduced to the minimum, *uClibc* or *Musl* if you are not yet using it for your root filesystem
- ▶ Reduce BusyBox to the strict minimum. You could even do without it and implement `/init` in C.
- ▶ Use statically linked applications (less CPU overhead, less libraries to load, smaller initramfs if no libraries at all), `BR2_STATIC_LIBS` in Buildroot.



Statically linked executables: licensing constraints

- ▶ Statically linked executables are very useful to reduce size (especially in small initramfs), and require less work to start.
- ▶ However, the LGPL license in the uClibc and glibc libraries requires to leave the user the ability to relink the executable with a modified version of the library.
- ▶ Solution to keep static binaries:
 - Either provide the executable source code (even proprietary), allowing to recompile it with a modified version of the library. That's what you do when you ship a static BusyBox.
 - Or also provide a dynamically linked version of the executable (in a separate way), allowing to use another library version.
 - Easiest solution: build your static executables with the `musl` library (MIT license: no trouble)
- ▶ References: <https://gnu.org/licenses/gpl-faq.html#LGPLStaticVsDynamic>
<https://gnu.org/copyleft/lesser.html#section4>

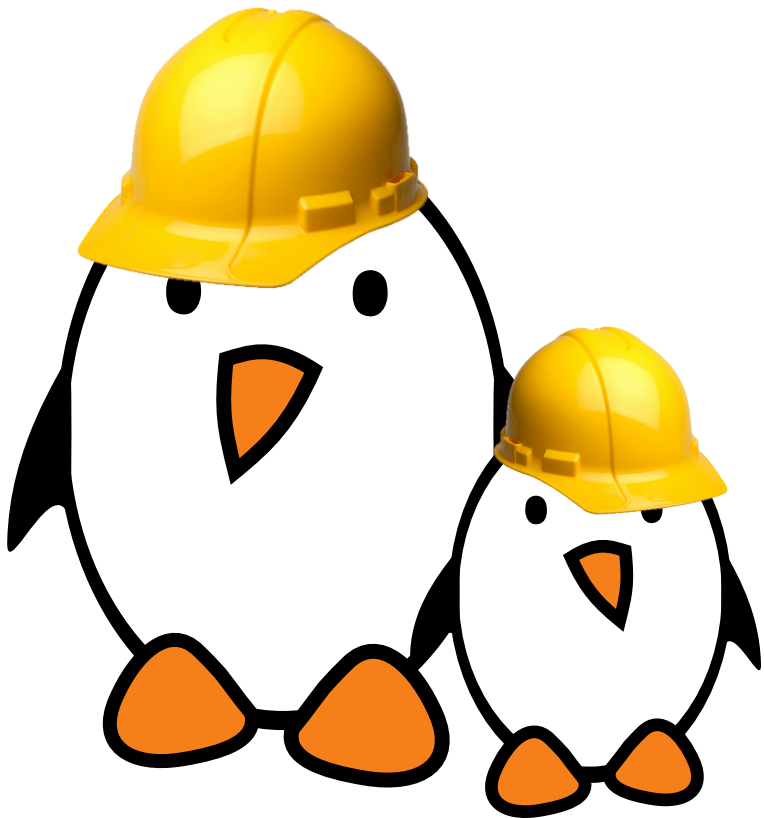


Do not compress your initramfs

- ▶ If you ship your initramfs inside a compressed kernel image, don't compress it (enable `CONFIG_INITRAMFS_COMPRESSION_NONE`).
- ▶ Otherwise, by default, your initramfs data will be compressed twice, and the kernel will be bigger and will take a little more time to load and uncompress.
- ▶ Example on Linux 5.1 with a 1.60 MB initramfs (tar archive size) on Beagle Bone Black: this allowed to reduce the kernel size from 4.94 MB to 4.74 MB (-200 KB) and save about 170 ms of boot time.



Practical lab - Filesystem optimizations



- ▶ Comparing the boot time performance of various filesystems
- ▶ Tests with initramfs booting too



Filesystem optimizations - Results

Results on BeagleBone Black, Linux 5.11

FS image size	Buildroot image size	zImage size diff	Time to init	Total boot time	ffmpeg exec time
ext2 (rev1) only	62,914,560	+19,544	8.489s	9.704s	0.498s
ext4 only	62,914,560	+241,472	8.645s	9.862s	0.484s
btrfs	114,294,784	+546,376	11.789s	12.918s	0.487s
f2fs	104,857,600	+167,640	8.670s	9.803s	0.488s
squashfs with lzo	724,992	+19,016	8.500s	9.721s	0.436s
erofs without compression	1,196,032	+28,072	8.510s	9.795s	0.491s
initramfs	N/A	+169,552	8.399s	9.660s	0.455s

Note: **zImage** kernel compressed with LZMA (best but slowest compressor), with a 1.162 MB filesystem (size of the tar archive generated by Buildroot).